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EXTENDED ABSTRACT

Integrating a ML-enhanced Radiation Parameterization into ICON-XPP

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1. Introduction and objectives

The radiation parameterization is a computational expensive component of Earth System Models (ESMs) and therefore has been target for computational efficient machine-learned surrogates [Chevallier(1998)]. In coarse-scale climate simulations (around 100 km horizontal resolution), the radiation parameterization is called less often than other parameterizations (1-2 h vs. 10 min) while keeping the cloud-radiative effects unchanged between radiation updates (Figure 1a). Increasing the radiation time step using a faster ML-based radiation emulator leads to fairly similar results because deficiencies in the representation of subgrid-scale clouds remain. Therefore, we utilize the knowledge from ML-based radiation emulation to build a ML-enhanced radiation parameterization that targets the improvement of radiative tendencies by implicitly including subgrid-scale cloud effects. The ML-enhanced parameterization is then integrated into the ICOsahedral Nonhydrostatic-Earth System Model (ICON-ESM).

2. Methods and data

We use high-resolution ICON simulations [Giorgetta(2022)] with a horizontal resolution of 5 km and coarse grain it to 80 km. Here, the cloud effect on heating rates is defined by the difference between all-sky and clear-sky heating rates. Coarse-grained high-resolution simulations implicitly include subgrid-scale cloud effects. The method is based on [Hafner(2025)], where a bidirectional long-short term memory (BiLSTM) predicts the cloud effects on radiative heating rates based on the coarse input variables. Here, we applied the same method to a 130 vertical level grid, and we coupled the ML-enhanced radiation scheme to ICON-XPP [Müller(2025)] using FTorch [Atkinson(2025)].

3. Preliminary results and/or expected outcomes

Offline results show 4-10 times smaller errors in tropospheric heating rates for different regimes and regions. The improvement is larger for longwave heating rates. This is promising for reduced biases for the cloud radiative effect, for example those associated with stratocumulus regions [Bock(2023)].

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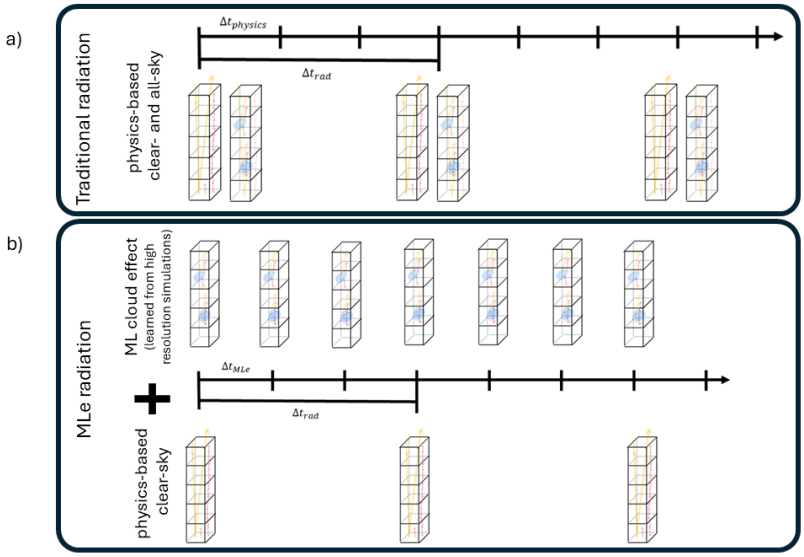

Figure 1. *Computation strategy of the radiation parameterization. a) The usual setup of radiation computes clear- and all-sky fluxes every radiation time step Δt_{rad}, which is a multiple of the physics time step $\Delta t_{physics}$. For every $\Delta t_{physics}$, the fluxes are adjusted, e.g. shortwave fluxes are scaled by the change in solar zenith angle. b) In the ML-enhanced setup, the clear-sky fluxes are calculated by the physics-based scheme and the cloud effects are computed by a neural network more frequently (every Δt_{MLe}), which can be the same as $\Delta t_{physics}$.*

The separation of cloud effects and clear-sky radiation has several advantages compared to a regular emulation. First, the clear-sky component will be computed by the physics-based radiation scheme, which allows the hybrid radiation scheme to be responsive to changes in greenhouse gases and aerosols. Second, the clear-sky part can be called less often, while cloud effects can be updated every time step. This way computational cost can be optimized (Figure 1b).

While the coupling works technically, ongoing work includes improving numerical stability, evaluation of online results and comparison to observations.

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